

How to Reproduce (Day 4 — Frozen Pipeline)

The full scientific pipeline (raw ERA5 → UTCI → NCTL → CTBI) is consolidated into a single script for reproducibility:

```
python src/pipeline_end_to_end.py
```

Input required: data/processed/era5_merged.nc (already in the repo). To regenerate this from scratch instead of using the version already committed, see src/download_era5.py and src/read_era5.py.

Outputs (all written to data/processed/):

File	Contents
utci_hourly.csv	Hourly UTCI, IST-corrected, all 9 grid points
daystress.csv	Daily mean daytime UTCI
nctl.csv	Per-night nighttime excess load
CTBI.csv	Cumulative Thermal Burden Index, $\alpha=0.7$, reset per year
ctbi_sensitivity.csv	CTBI at $\alpha=0.6 / 0.7 / 0.8$, for sensitivity comparison

This script reuses the team's own already-validated logic (process_weather.py, cos_solar.py, mrt.py, utci.py) rather than reimplementing it, so it can't drift out of sync with what's already been checked. It incorporates all fixes made during Day 3:

1. UTC → IST timezone conversion, applied before any day/night hour classification
2. NCTL computed per single night (no carry-forward accumulation between nights)
3. Nighttime baseline = fixed 90th percentile (not a median/expanding-window baseline)
4. Normalization = $\text{min-max} (x - \text{min}) / (\text{max} - \text{min})$, not divide-by-max
5. CTBI resets to 0 at the start of each year (2021–2026 are 6 separate seasons, not one continuous series)

Verified reproducible: running this script from a clean environment regenerates a baseline of 32.15°C and a CTBI range of 0.089–0.754 — matching the values already in the repo exactly.

Pipeline is frozen for Day 4

No new scientific features are being added to this pipeline going forward. Any further Day 4+ work builds *on top of* these outputs (spatial join, vulnerability, priority engine, dashboard) rather than modifying the core UTCI/NCTL/CTBI logic itself. Bug fixes only, if any are found.